```
06 eb 02 9d 7f ff 01 68 62 81 69
```

Even identical sensors might output differently based on firmware versions or configuration. My sensor deployment uses 10 distinct data formats, each requiring specific handling. This necessitates flexible integration and data modeling approaches.

VERACITY

The accuracy and trustworthiness of IoT data can be inconsistent, subject to sensor errors, transmission issues, or device failures. Environmental factors affect sensor accuracy—my outdoor temperature sensors read high in direct sunlight. Occasionally my indoor temperature sensor produces an impossible value of -99°C indicating a malfunction that I must handle.

VALUE

While the ultimate goal is to extract meaningful insights and actionable intelligence from raw data the value can vary dramatically. My soil moisture sensors show noise-level fluctuations between actual watering events. This low value density means processing must extract signal from noise, identifying genuinely significant changes while filtering meaningless variation.

3.5.2 Message decoding

Sensors transmit data in different formats, using different encoding and protocols that must be decoded and transformed into a format that can be ingested, stored, and processed by a digital twin application. This requires some compute to run to perform the decoding. Since the volume and velocity of sensor data can he highly variable, it makes sense to use serverless compute for message decoding. In my home digital twin I use AWS Lambda functions that activate only when messages arrive, eliminating idle compute costs. For my home digital twin processing approximately 50,000 messages monthly, this translates to mere dollars in compute costs versus hundreds for an always-on server to do the same.